CHAPTER 5

Signal Encoding Techniques

“Thus one says, in general, that the function of the transmitter is to encode, and that of the receiver to decode, the message. The theory provides for very sophisticated transmitters and receivers—such, for example, as possess ‘memories,’ so that the way they encode a certain symbol of the message depends not only upon this one symbol but also upon previous symbols of the message and the way they have been encoded.”

—*The Mathematics of Communication*,
Scientific American, July 1949,
Warren Weaver

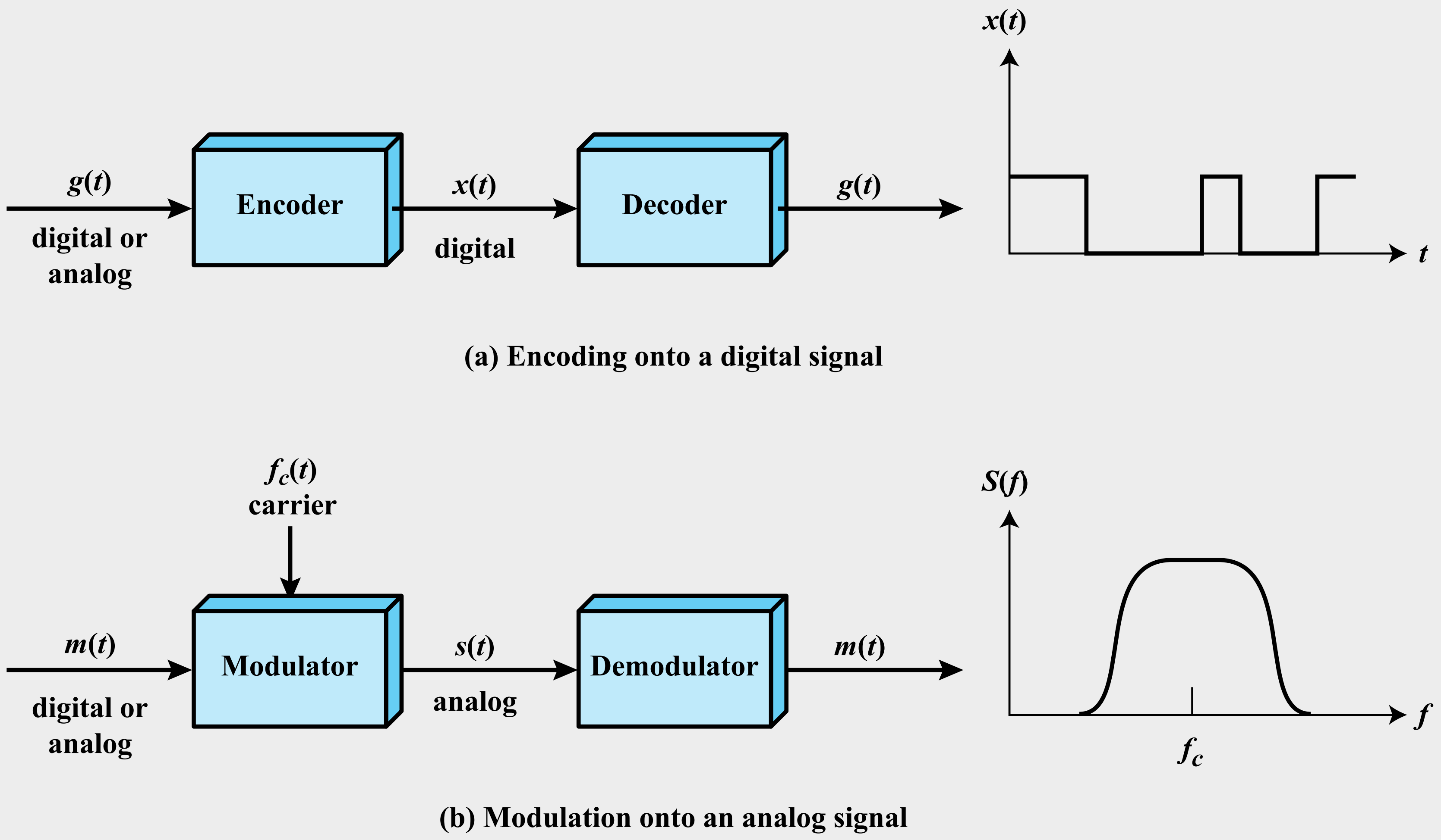


Figure 5.1 Encoding and Modulation Techniques

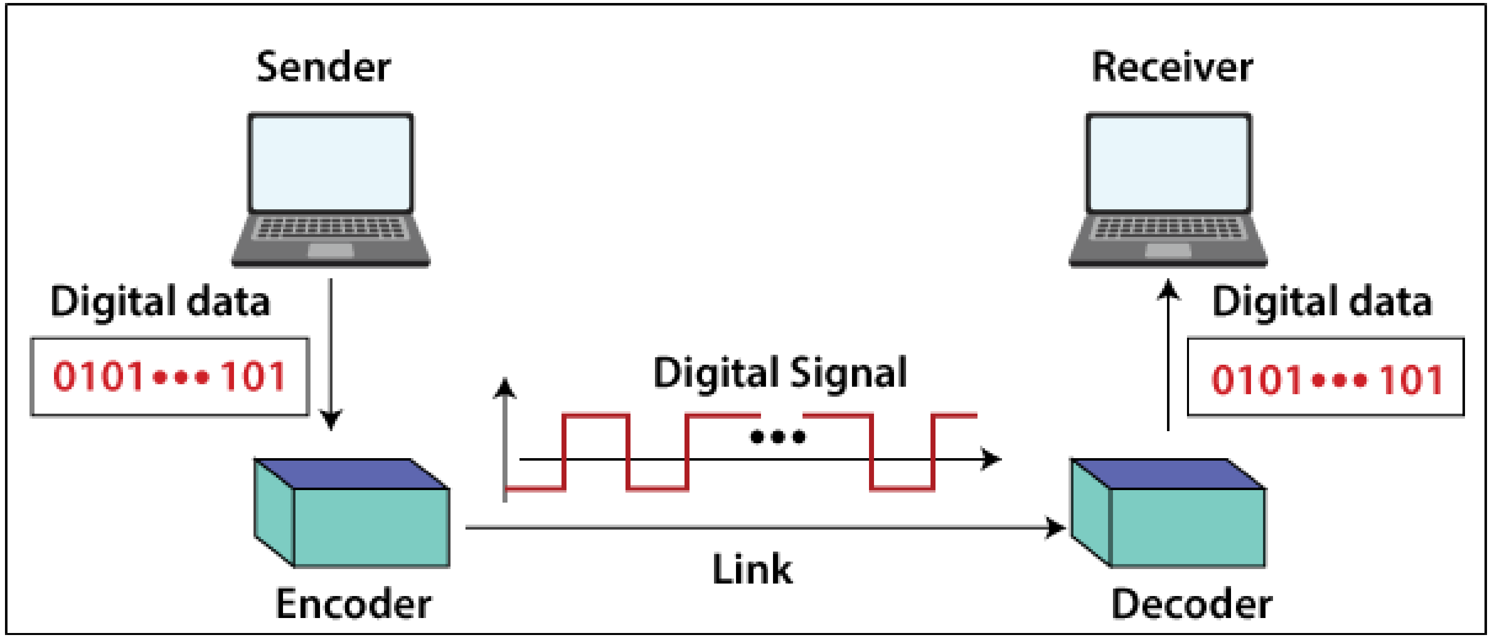
- Digital data, digital signal
- Digital data, analog signal
- Analog data, digital signal
- Analog data, analog signal

Digital Data, Digital Signal

- Digital signal
  - Sequence of discrete voltage pulses
  - Each pulse is a signal element
- Binary data are transmitted by encoding each data bit into signal elements

Line coding and decoding

Converting a digital data of 1’s and 0’s into a sequence of signals (voltage level (+V) and a low voltage level (0 or -V)).



Key Data Transmission Terms

Term	Units	Definition
Data element	Bits	A single binary one or zero
Data rate	Bits per second (bps)	The rate at which data elements are transmitted
Signal element	Digital: a voltage pulse of constant amplitude Analog: a pulse of constant frequency, phase, and amplitude	That part of a signal that occupies the shortest interval of a signaling code
Signaling rate or modulation rate	Signal elements per second (baud)	The rate at which signal elements are transmitted

Relationship between data rate and signal rate

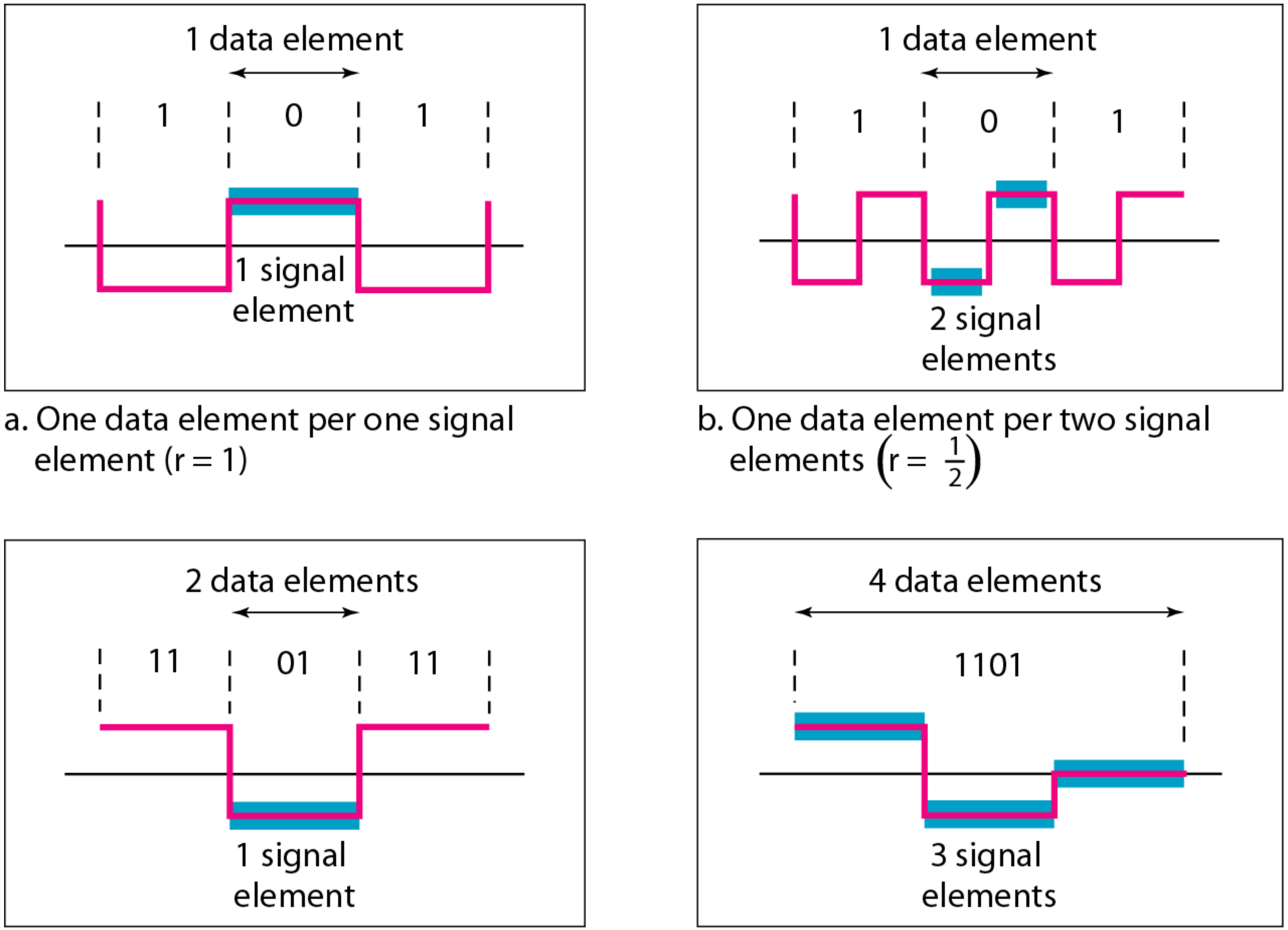
- The data rate defines the number of bits sent per sec - bps.
- The signal rate is the number of signal elements sent in a second and is measured in bauds.

- Bit rate = (number of bits per pulse) x (baud rate)

Mapping Data symbols onto Signal levels

- A data symbol (or element) can consist of a number of data bits:
  - 1 , 0 or
  - 11, 10, 01, ……
- A data symbol can be coded into a single signal element or multiple signal elements
  - 1 -> +V, 0 -> -V
  - 1 -> +V and -V, 0 -> -V and +V
- The ratio ‘r’ is the number of data elements carried by a signal element.

Signal element versus data element



a. One data element per one signal element ($r = 1$)

b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$

c. Two data elements per one signal element ($r = 2$)

d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

Nonreturn to Zero-Level (NRZ-L)
0 = high level
1 = low level

Nonreturn to Zero Inverted (NRZI)
0 = no transition at beginning of interval (one bit time)
1 = transition at beginning of interval

Bipolar-AMI
0 = no line signal
1 = positive or negative level, alternating for successive ones

Pseudoternary
0 = positive or negative level, alternating for successive zeros
1 = no line signal

Manchester
0 = transition from high to low in middle of interval
1 = transition from low to high in middle of interval

Differential Manchester
Always a transition in middle of interval
0 = transition at beginning of interval
1 = no transition at beginning of interval

B8ZS
Same as bipolar AMI, except that any string of eight zeros is replaced by a string with two code violations

HDB3
Same as bipolar AMI, except that any string of four zeros is replaced by a string with one code violation

Definition of Digital Signal Encoding Formats

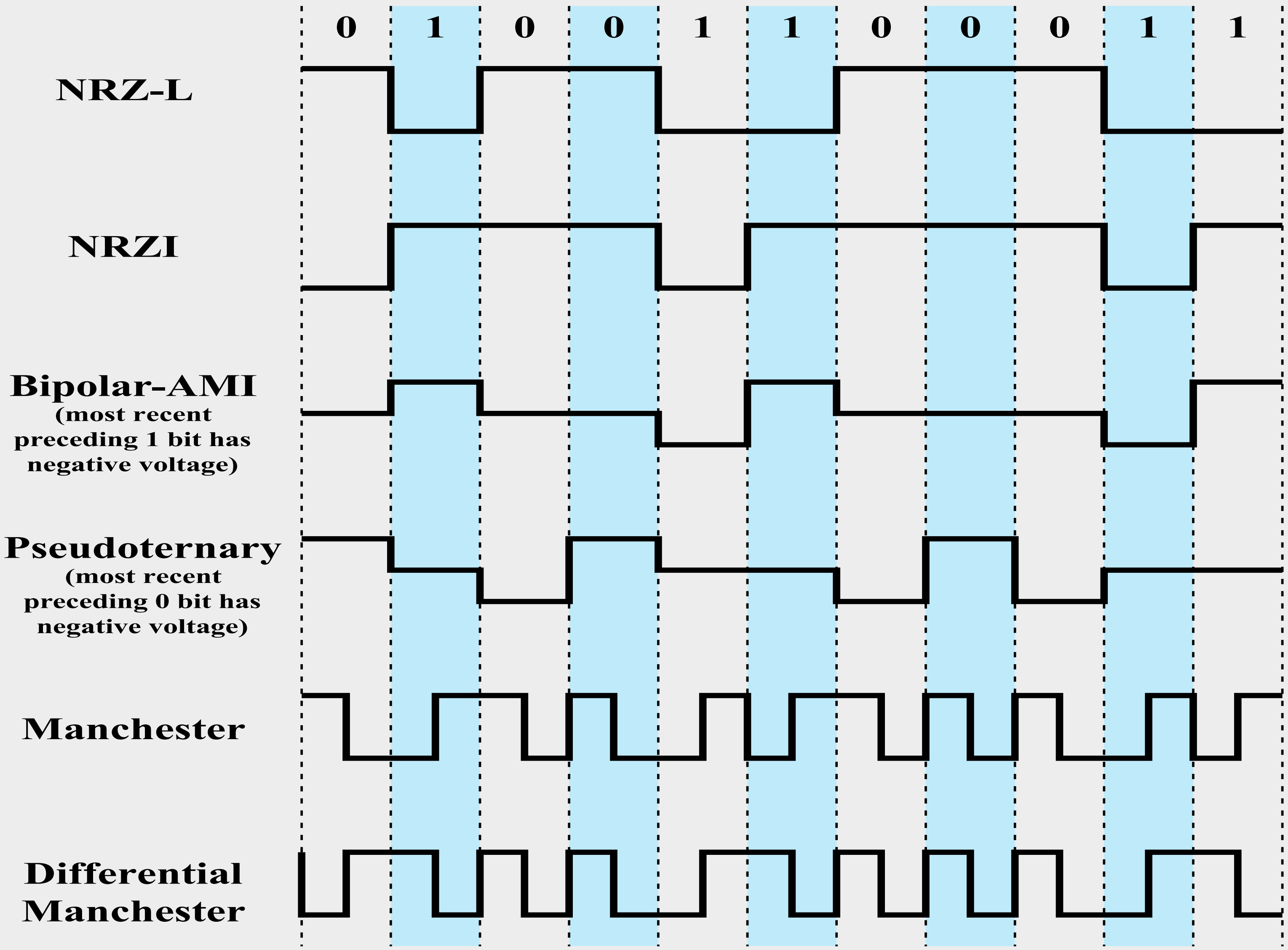
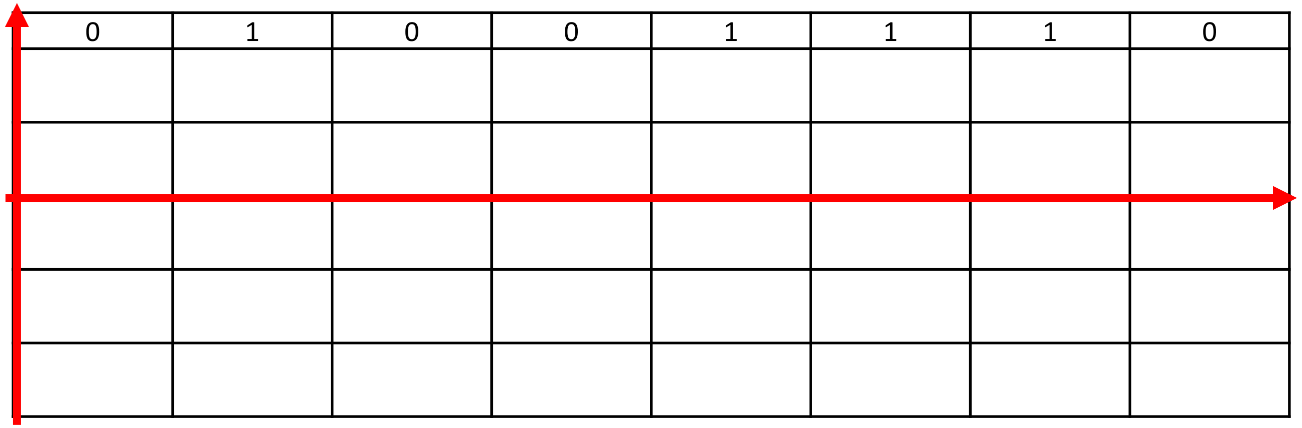


Figure 5.2 Digital Signal Encoding Formats

Nonreturn to Zero

- Voltage level is constant during a bit interval
  - No return to a zero-voltage level
  - Negative voltage represents 1 and a positive voltage represents 0



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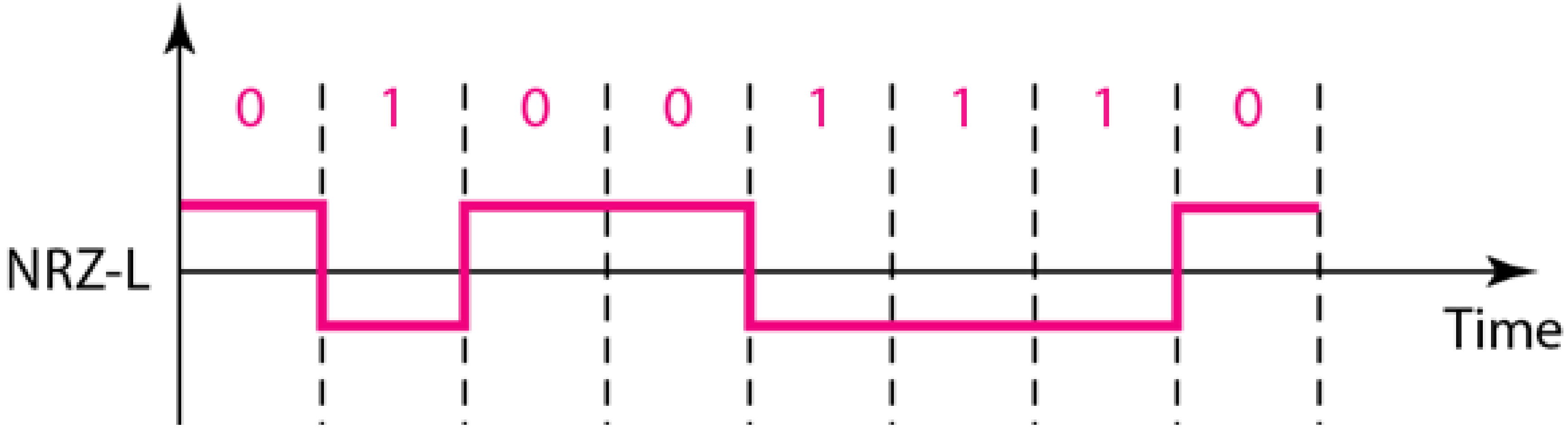
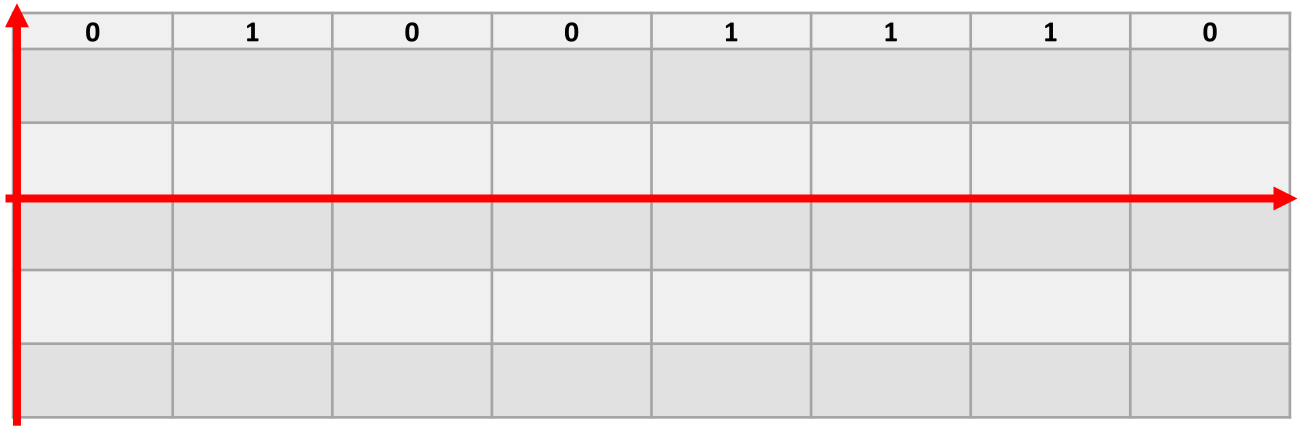


Non-return to Zero Inverted (NRZI)

- Non-return to zero, invert on ones
  - Transition (low to high, high to low) denotes binary 1
  - No transition denotes binary 0



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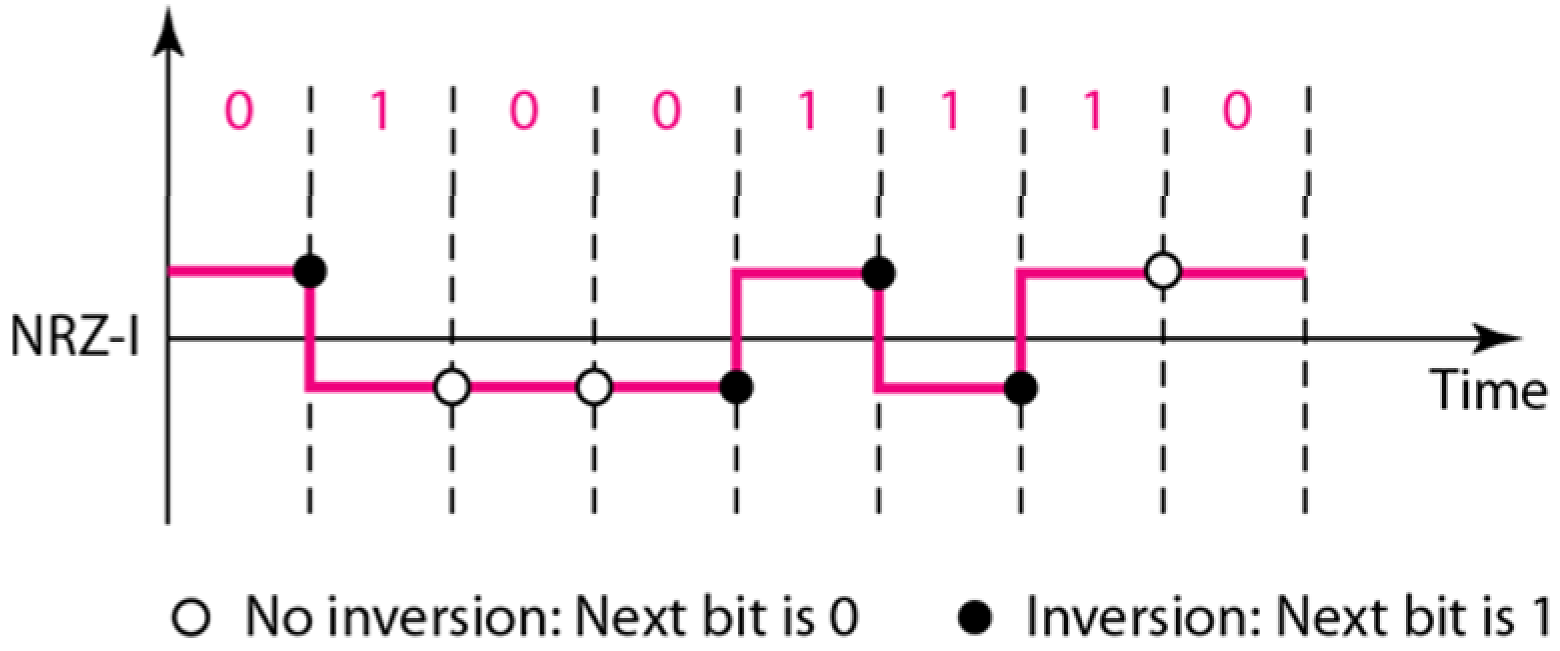
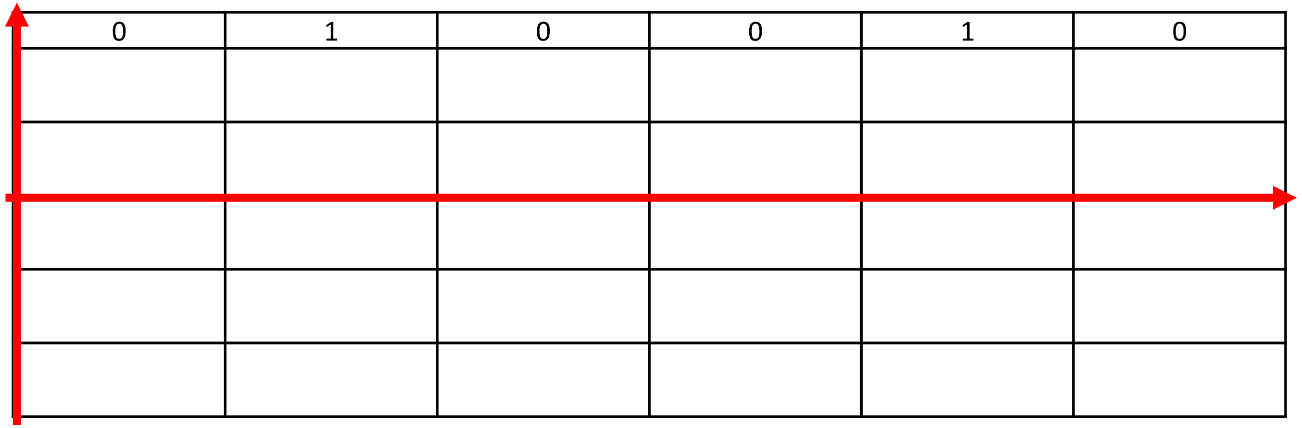


Multilevel Binary: Bipolar-AMI

- Binary 0 represented by no line signal
- Binary 1 represented by positive or negative pulse
- Binary 1 pulses alternate in polarity



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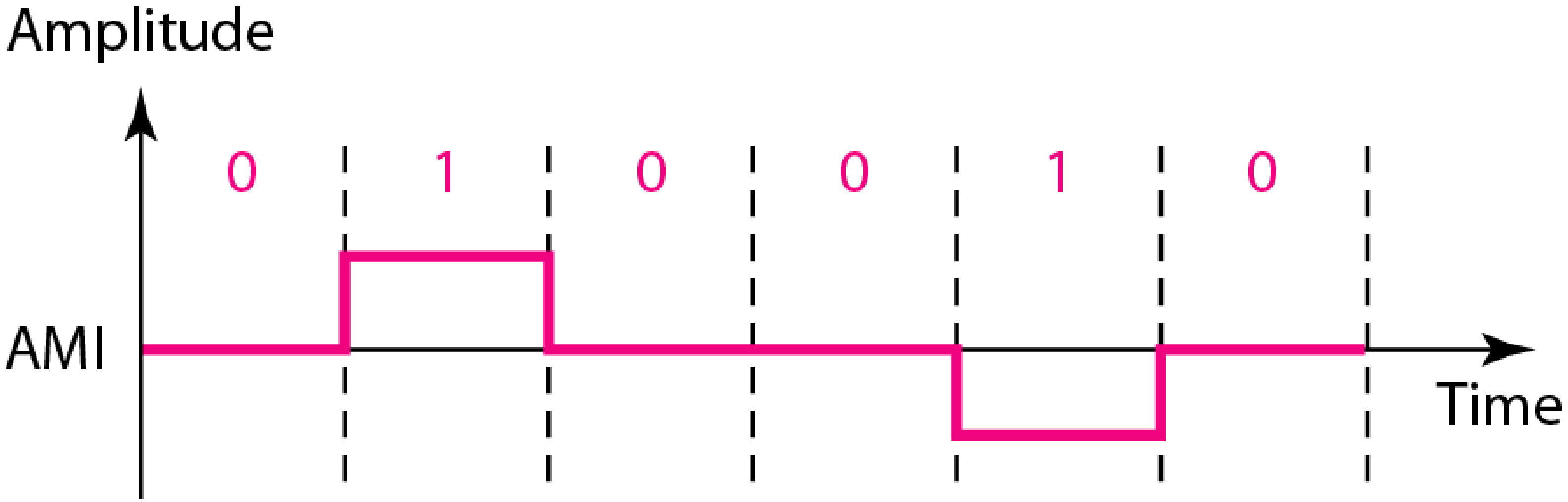
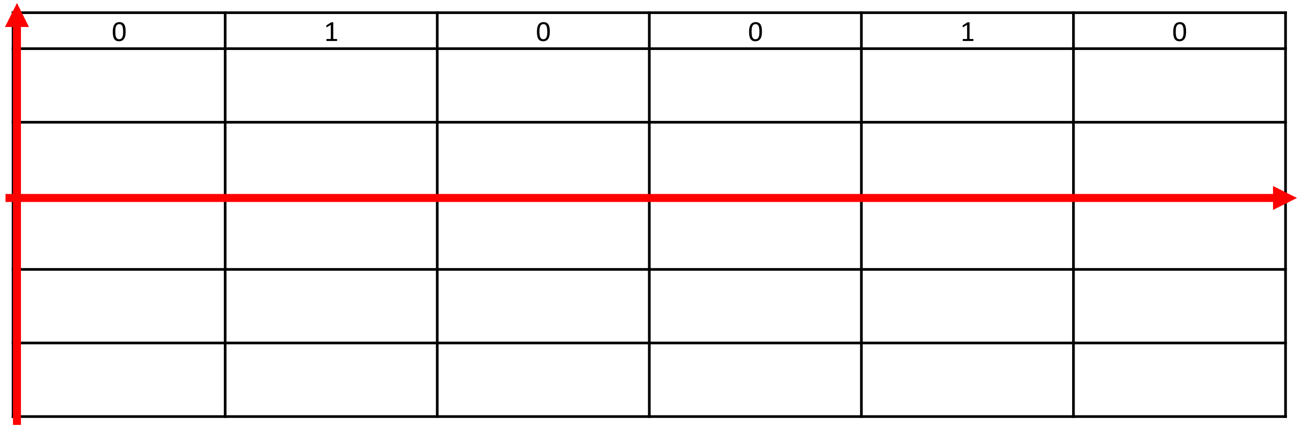


Multilevel Binary: Pseudoternary

- Binary 1 represented by absence of line signal
- Binary 0 represented by alternating positive and negative pulses



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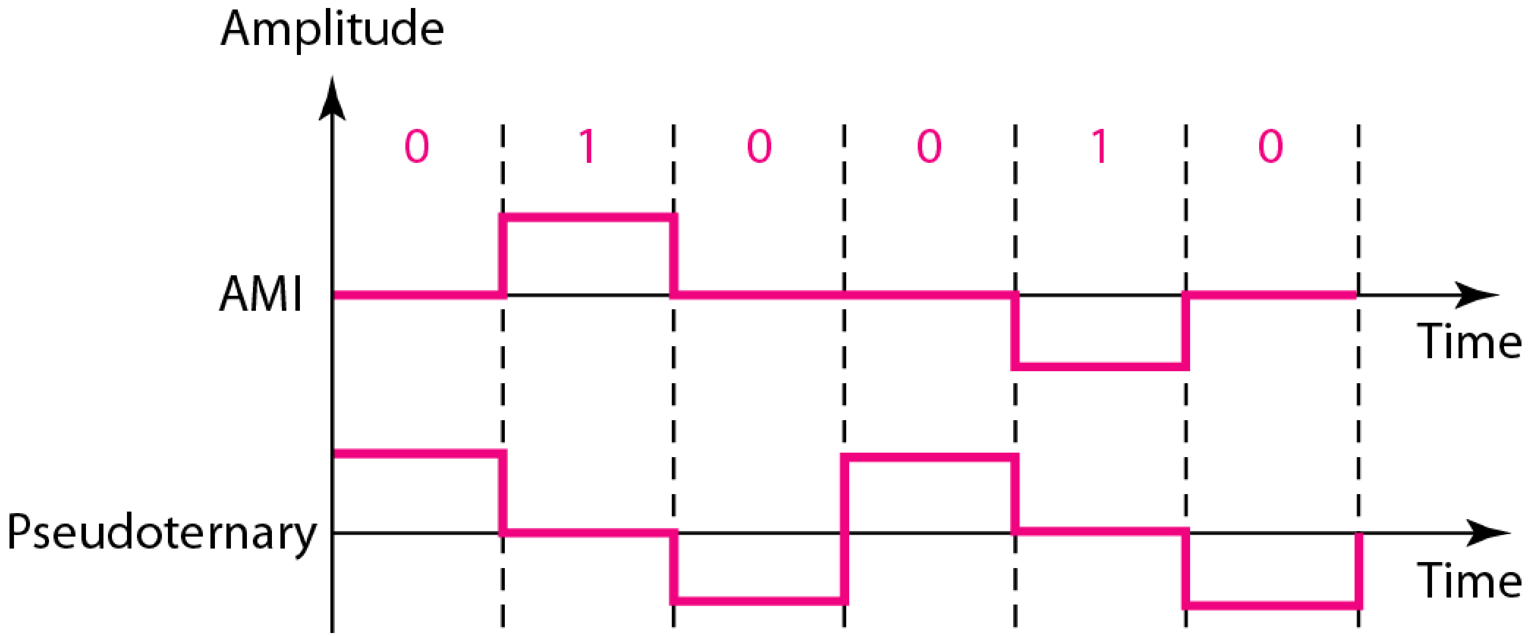
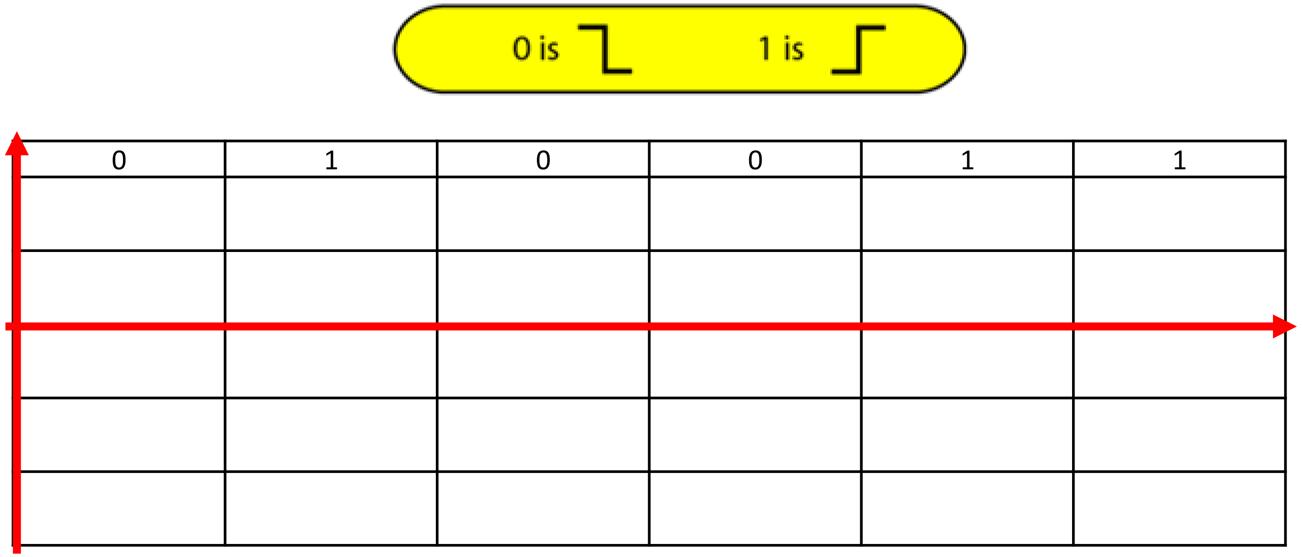


Manchester Encoding

There is a transition at the middle of each bit period

Low to high transition represents a 1

High to low transition represents a 0



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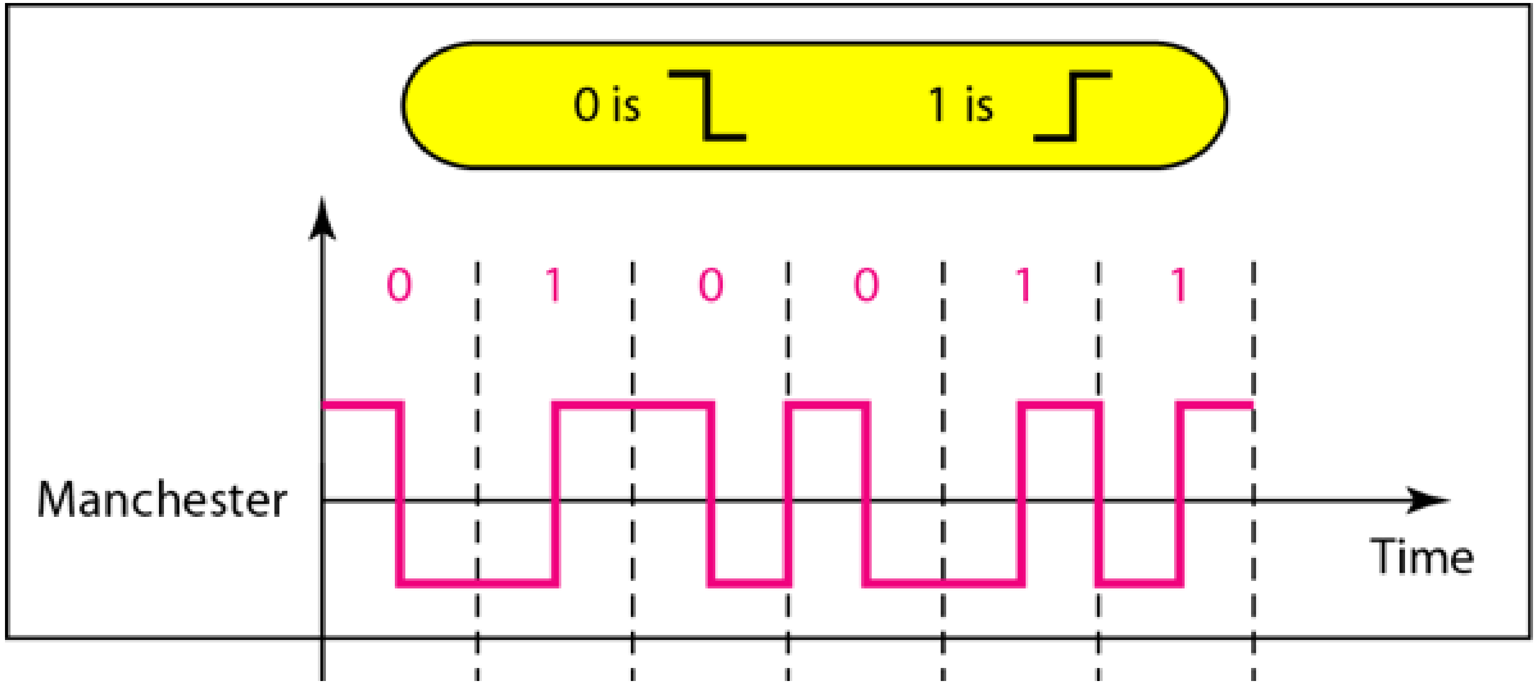
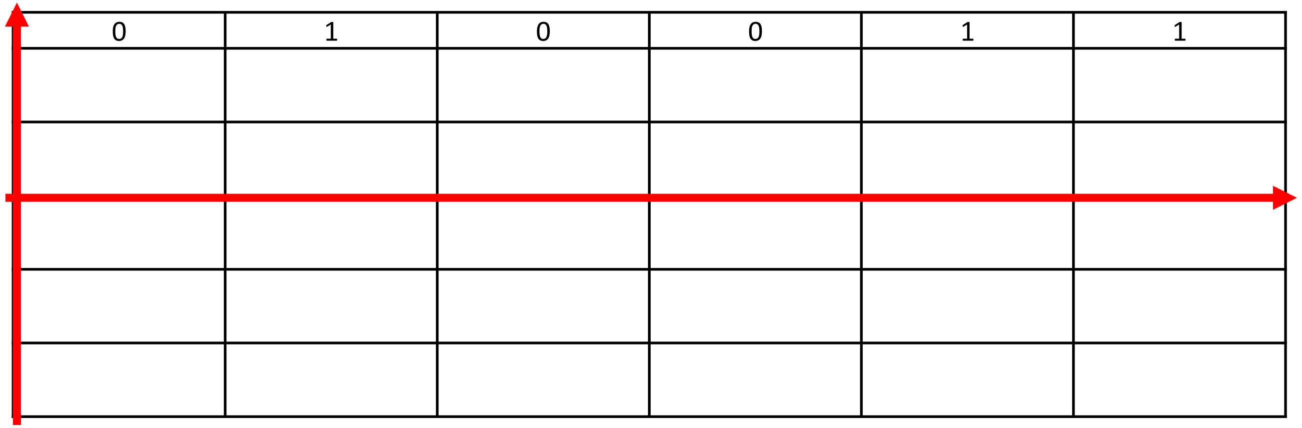


Differential Manchester Encoding

- The encoding of a 0 is represented by the presence of a transition at the beginning of a bit period
- A 1 is represented by the absence of a transition at the beginning of a bit period



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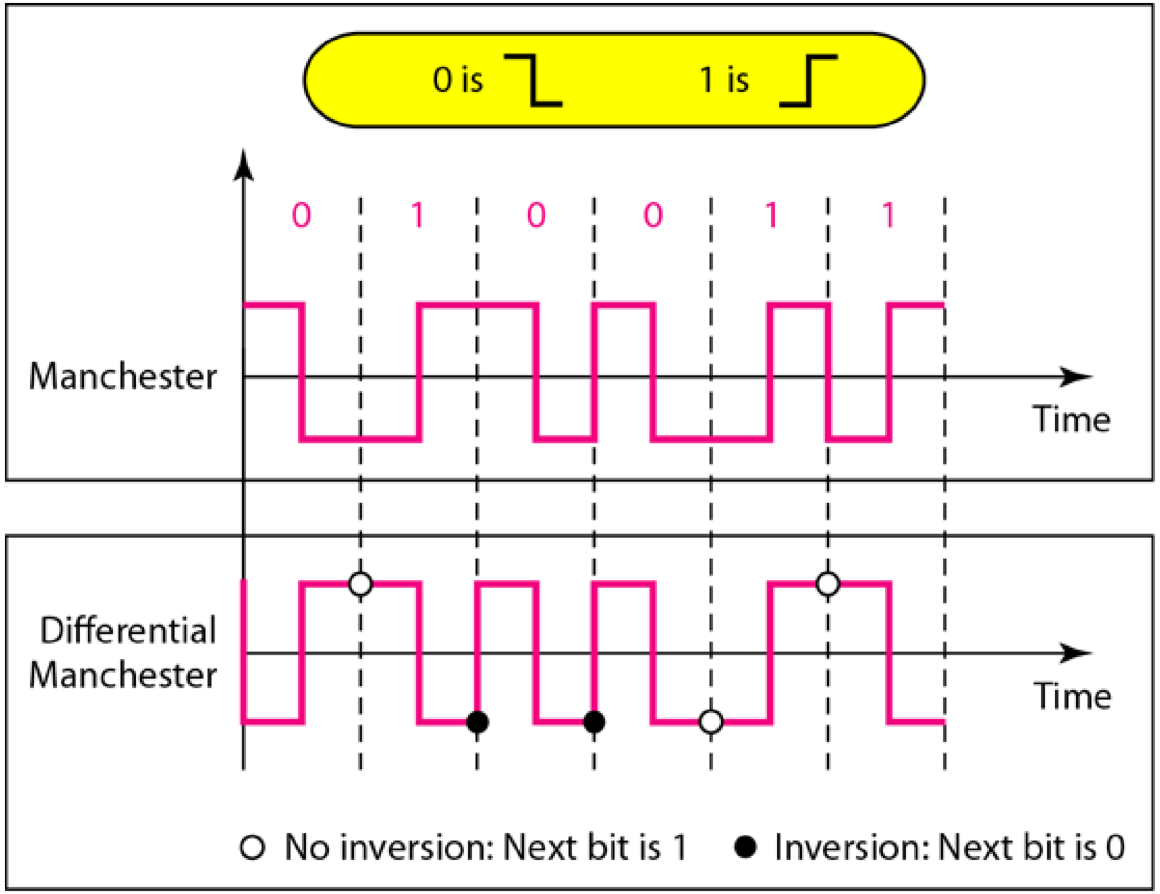


Digital Data, Analog Signal

- Digital data needs to be carried on an analog signal.
  - A carrier signal (frequency f_c) performs the function of transporting the digital data in an analog waveform.
- Uses modem (modulator-demodulator) to convert digital data to analog signals and vice versa

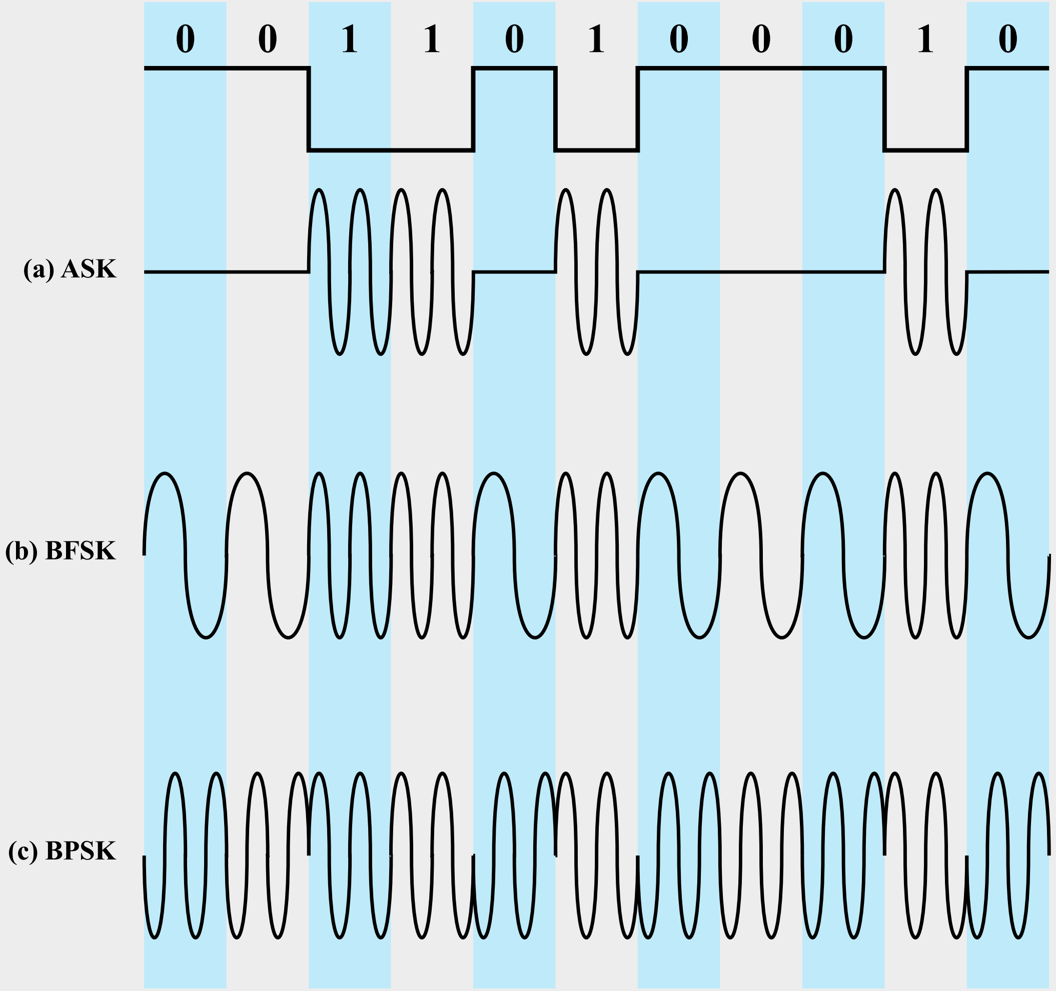
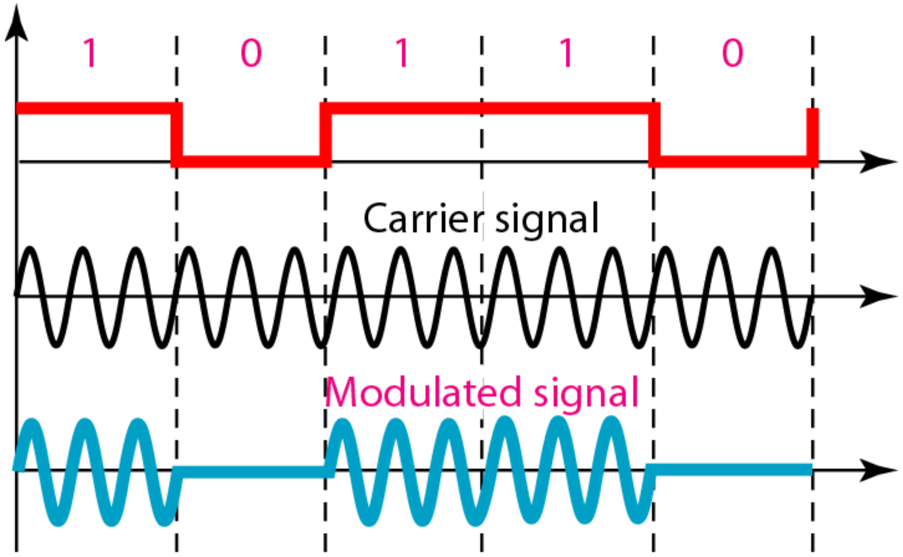
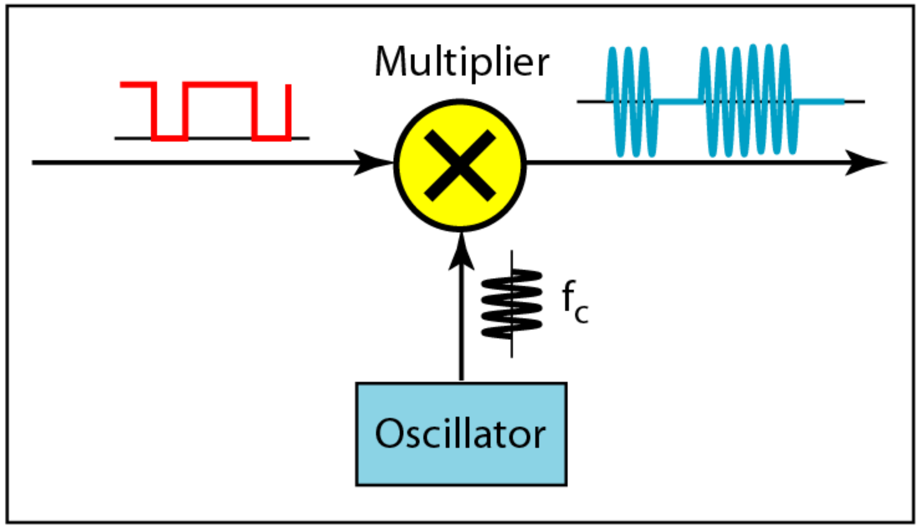


Figure 5.7 Modulation of Analog Signals for Digital Data

Amplitude Shift Keying (ASK)

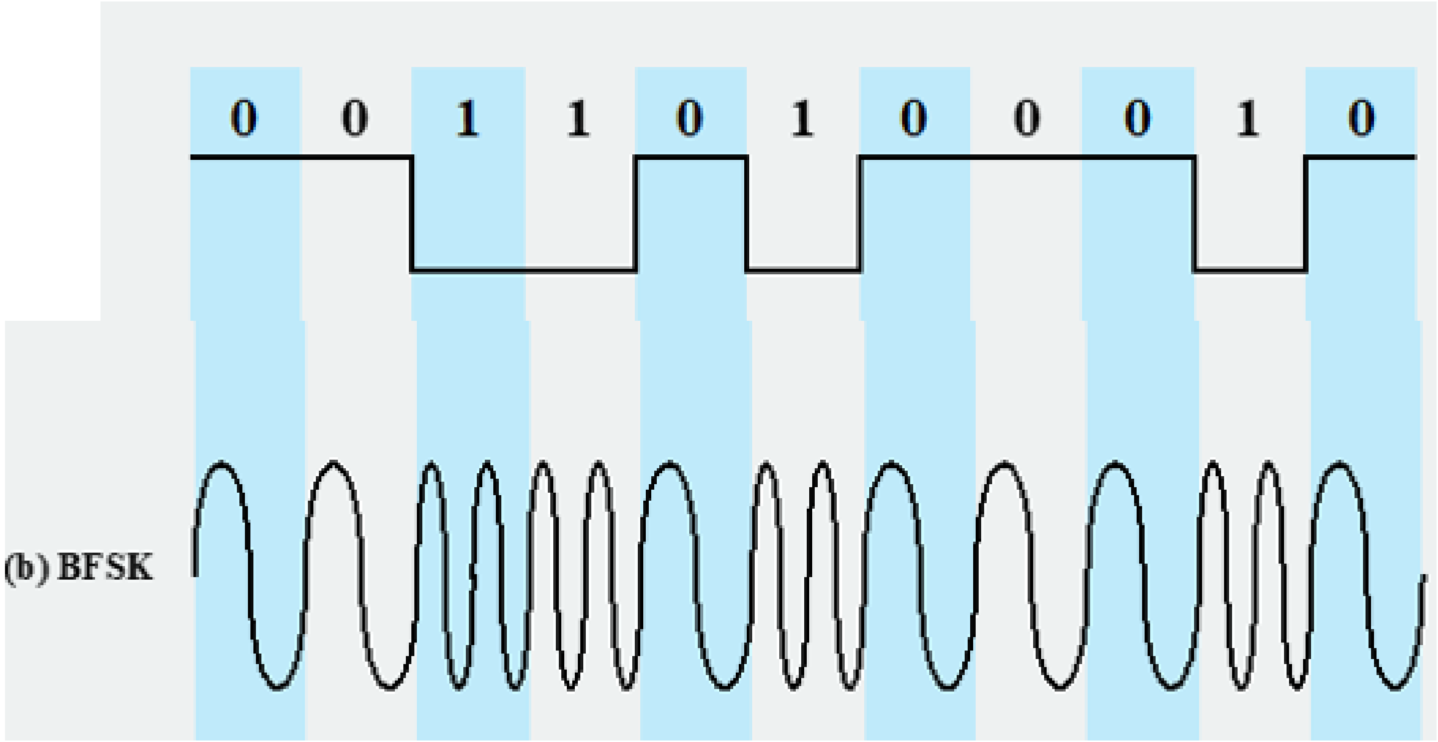
- ASK is implemented by changing the amplitude of a carrier signal to reflect amplitude levels in the digital signal.
- Encode 0/1 by different carrier amplitudes





Binary Frequency Shift Keying (BFSK)

- Two binary values are represented by two different frequencies



Phase Shift Keying (PSK)

- The phase of the carrier signal is shifted to represent data
- Binary PSK
  - Two phases represent the two binary digits
- Differential PSK
  - Phase shifted relative to previous transmission rather than some reference signal

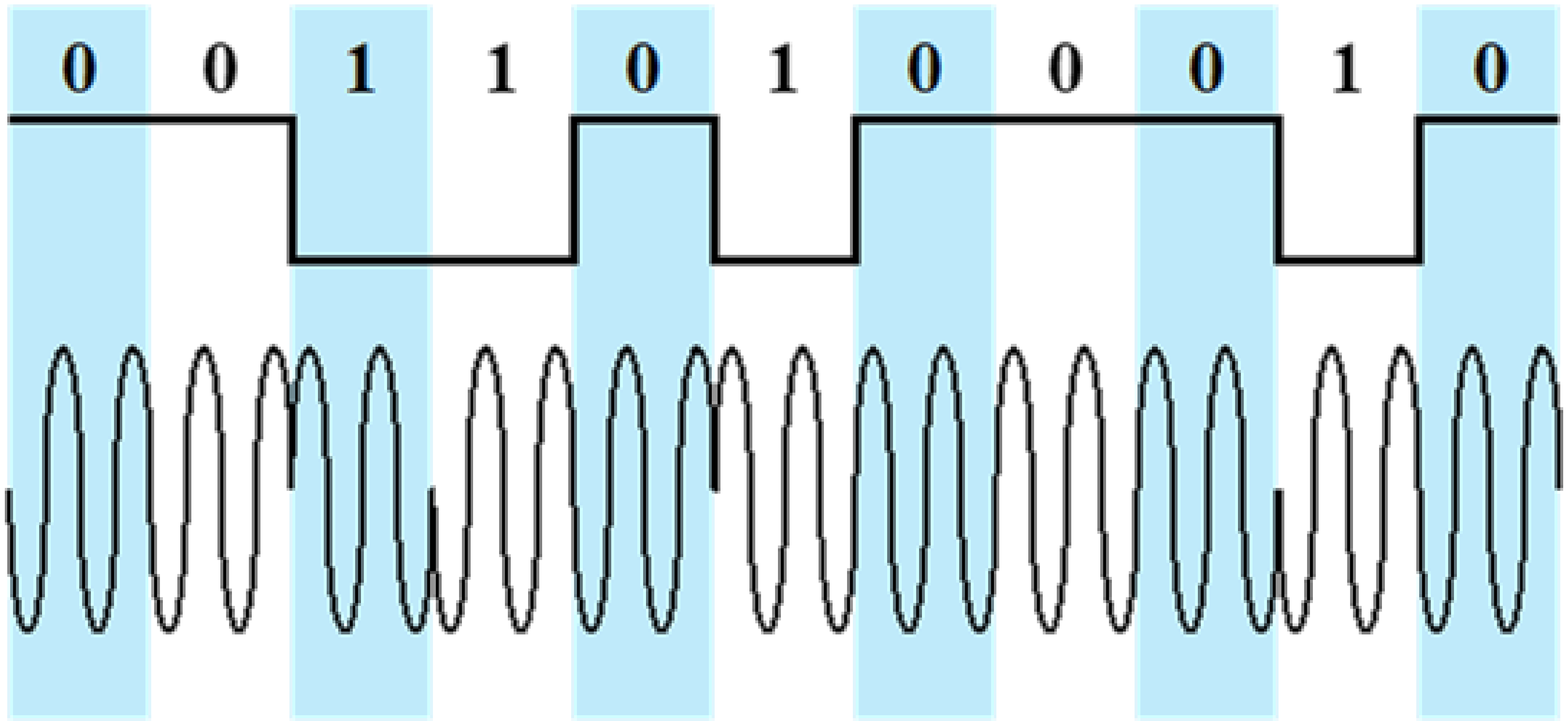


Figure 5.10 Differential Phase-Shift Keying (DPSK)

Analog Data, Digital Signal

- Digitization is the conversion of analog data into digital data
- Analog to digital conversion is done using a codec
  - Pulse code modulation
  - Delta modulation

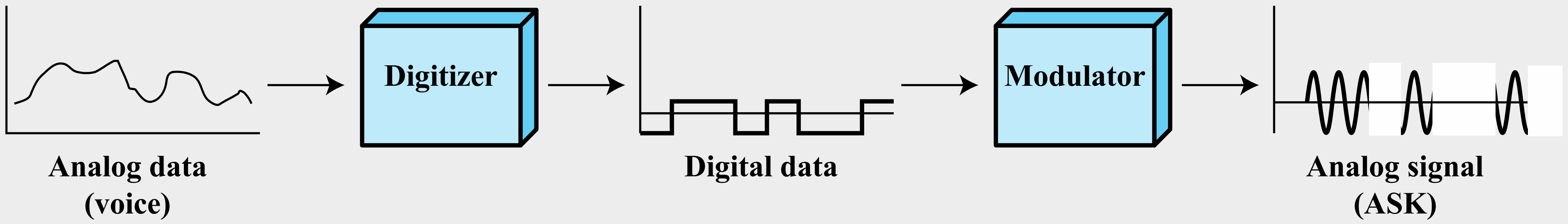


Figure 5.16 Digitizing Analog Data

Pulse Code Modulation (PCM)

- PCM consists of three steps to digitize an analog signal:
    1. Sampling
    2. Quantization
    3. Binary encoding
- Before we sample, we have to filter the signal to remove high frequency components that affect the signal shape.

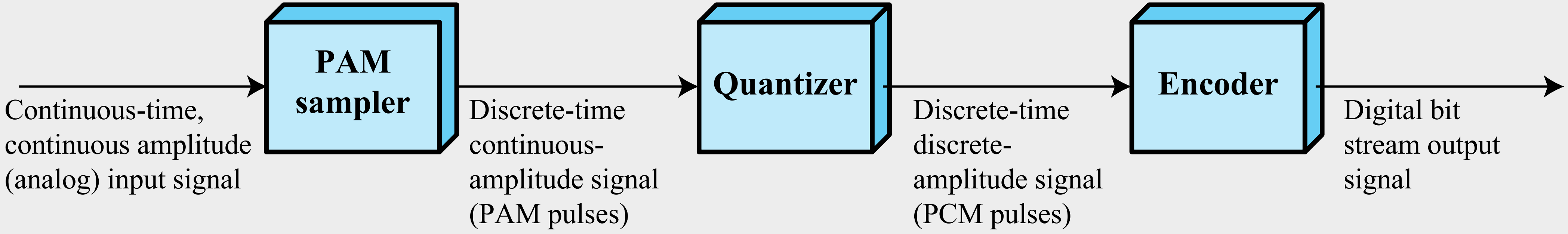
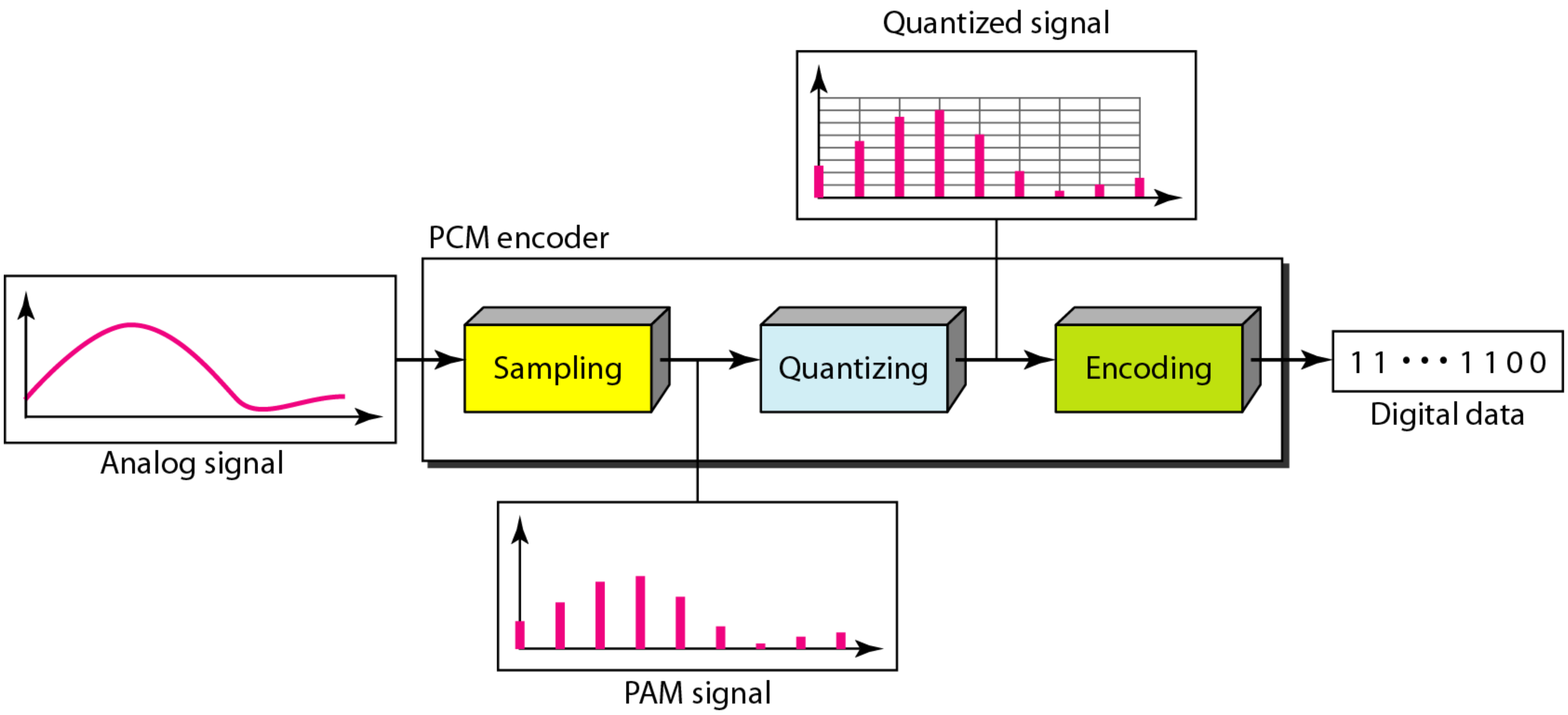


Figure 5.18 PCM Block Diagram

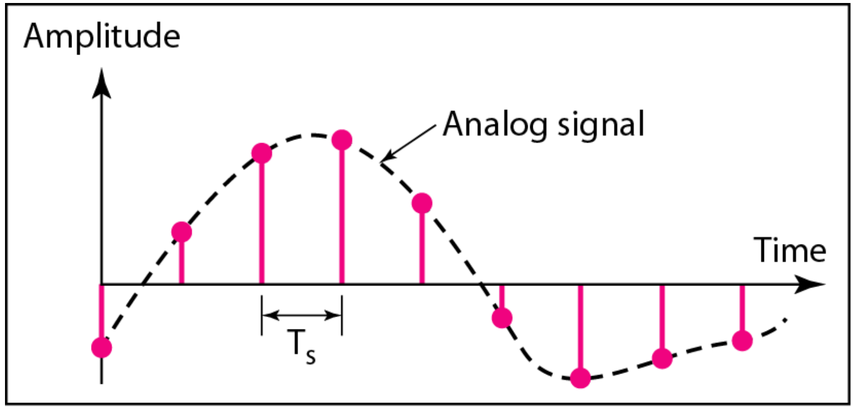
Components of PCM Encoder



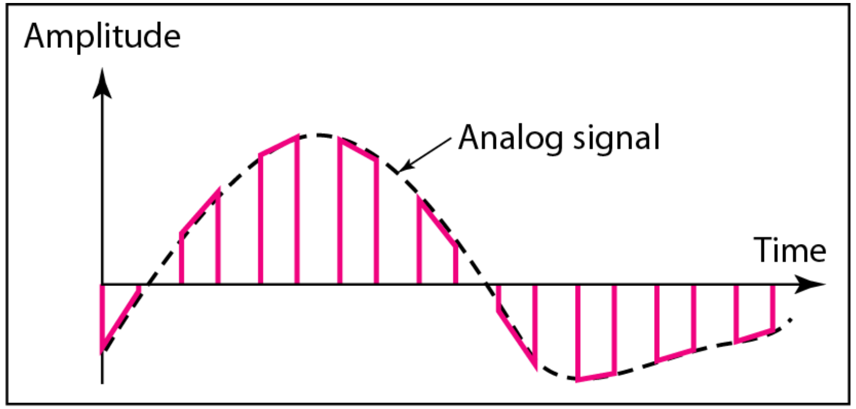
Three different sampling methods for PCM

- Pulse Amplitude Modulation (PAM)

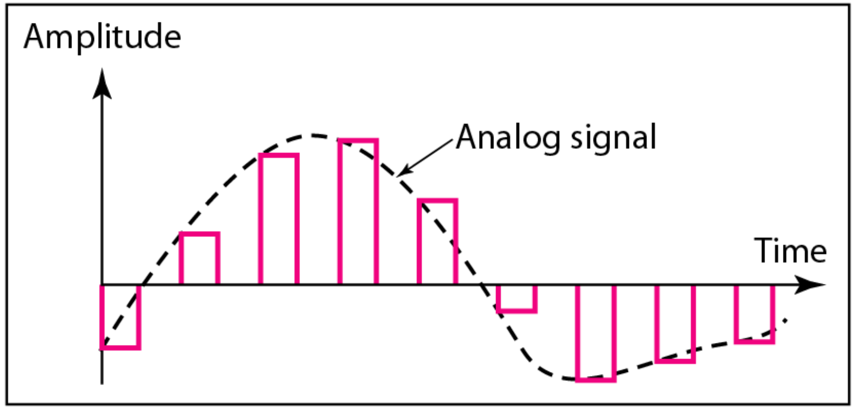
  To convert to digital, each of these analog samples must be assigned a binary code



a. Ideal sampling



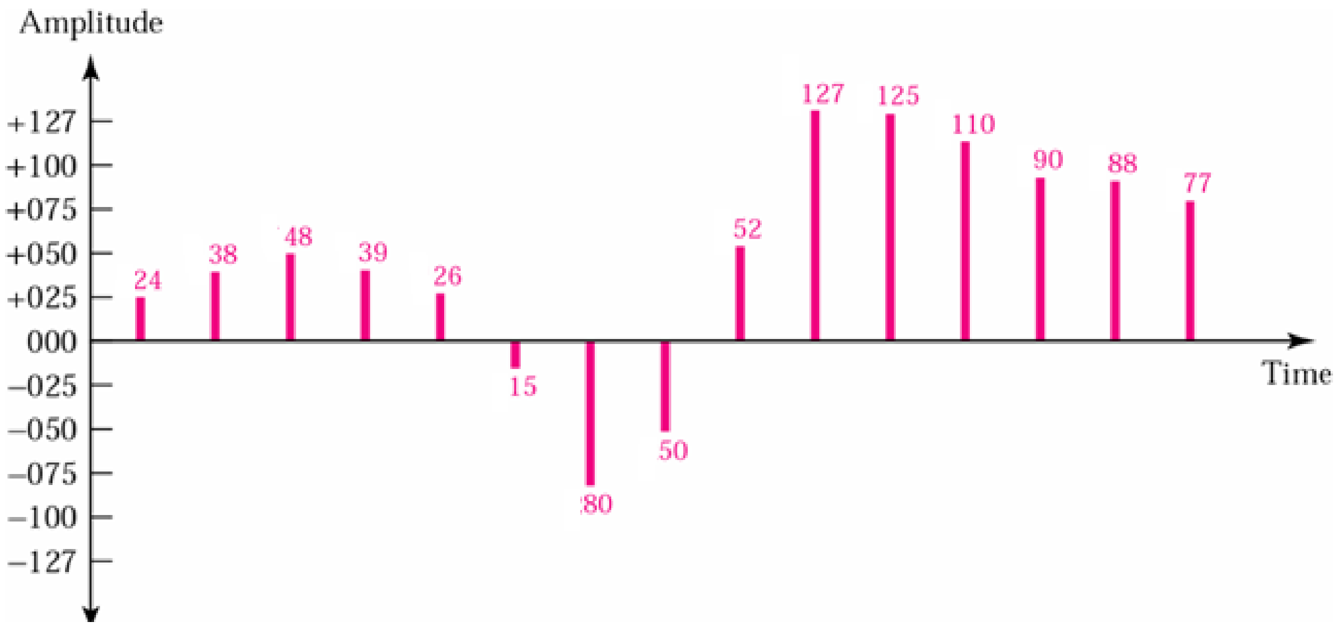
b. Natural sampling



c. Flat-top sampling

Quantized PAM Signal

- The result of PAM is a series of pulses with amplitude values between the maximum and minimum amplitudes of the signal with real values.
- **Quantization:** is a method of assigning integer values in a specific range to sampled instances.

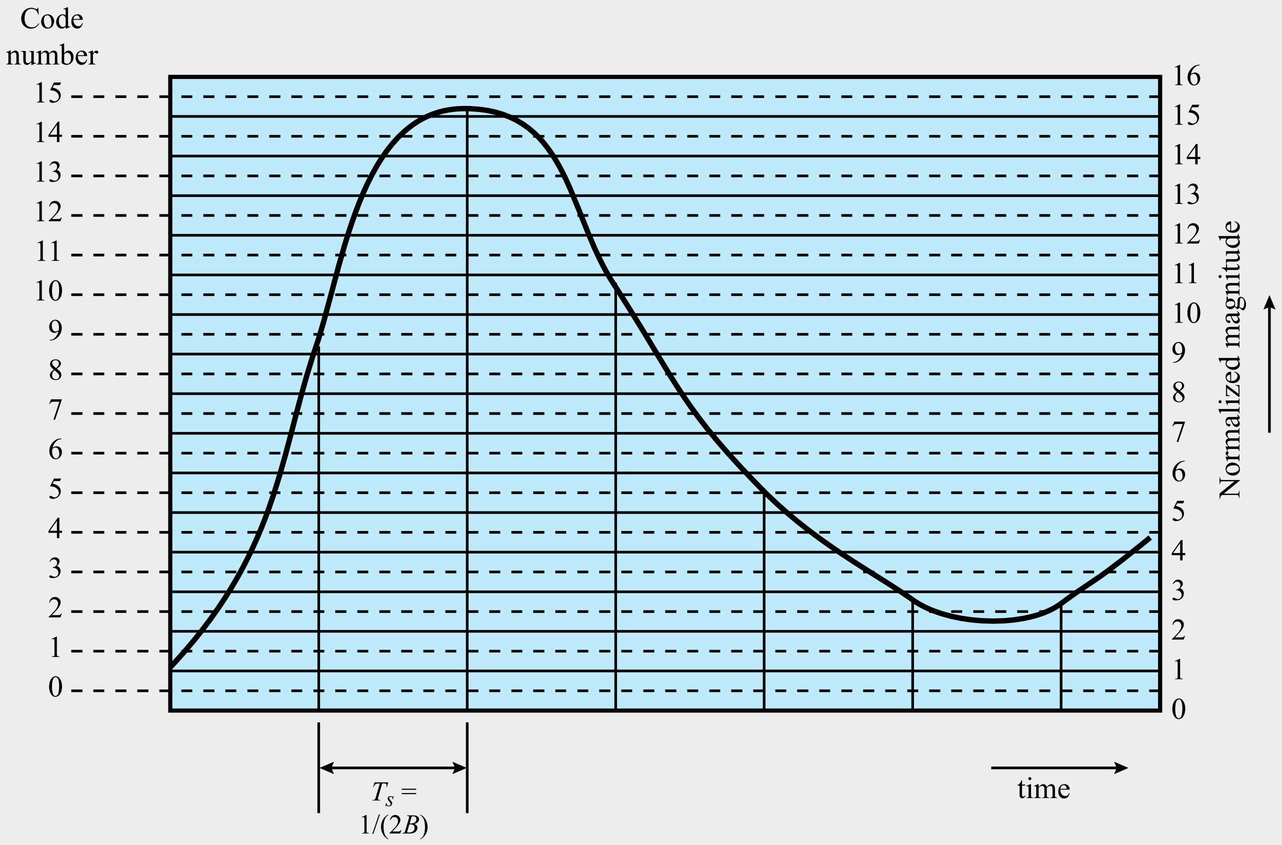


Binary Encoding

Each quantized samples is translated into equivalent binary codes.

+024	00011000	–015	10001111	+125	01111101
+038	00100110	–080	11010000	+110	01101110
+048	00110000	–050	10110010	+090	01011010
+039	00100111	+052	00110110	+088	01011000
+026	00011010	+127	01111111	+077	01001101

Sign bit
+ is 0 – is 1



PAM value	1.1	9.2	15.2	10.8	5.6	2.8	2.7
quantized code number	1	9	15	10	5	2	2
PCM code	0001	1001	1111	1010	0101	0010	0010

Figure 5.17 Pulse-Code Modulation Example

Delta Modulation (DM)

- Analog input is approximated by a staircase function
- Move up or down one level (δ) at each sample interval
- Binary behavior
  - Function moves up or down at each sample interval
  - Moving up: generating 1
  - Moving down: generating 0

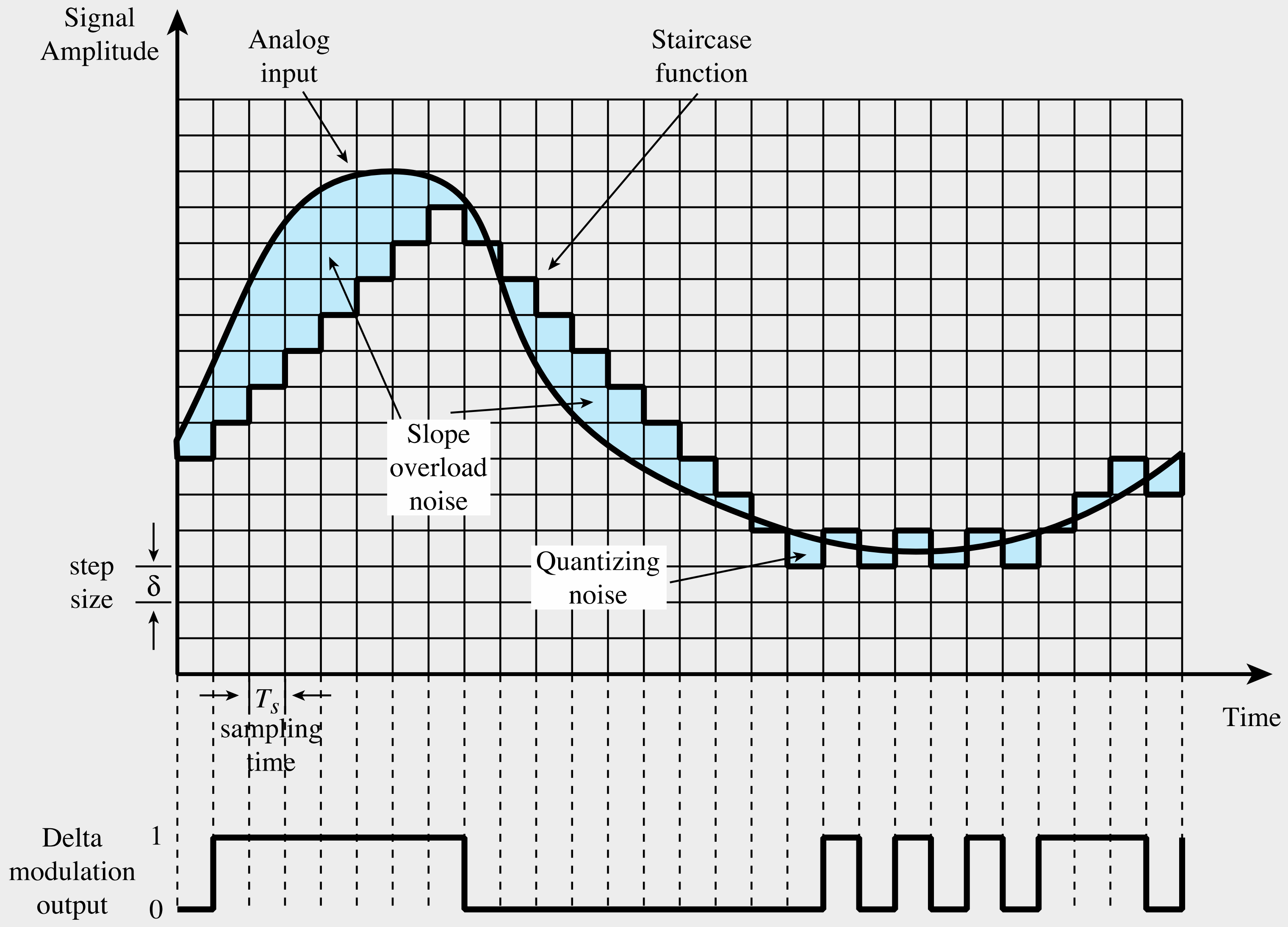


Figure 5.21 Example of Delta Modulation